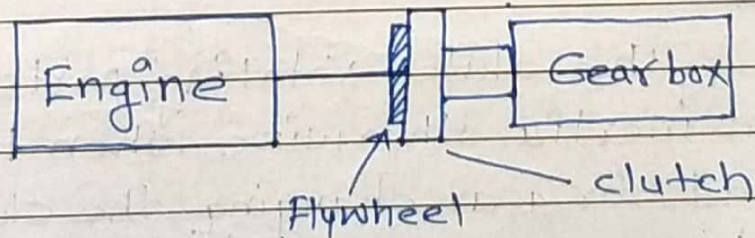


2.1 - Clutch

Introduction -

It is a mechanism which enables the rotary of motion one shaft to be transmitted when desired (at the time of working), to a second shaft the axis of which is co-insident with that of the first.

clutch is a device used to engage or disengage the transmission (i.e. gear box) from the engine.



* Requirement of clutch -

1. Torque transmission -

The clutch should be able to transmit the maximum torque of the engine under all condition. It is usually designed to transmit

125 to 150 % of the maximum engine torque.

2. Gradual Engagement -

The clutch should positively take the drive gradually without the occurrence of sudden jerk.

3. Heat dissipation -

During clutch application, large amount of heat are generated. The rubbing surface should have sufficient area and mass to absorb the heat generated.

4. Vibration damping -

Suitable mechanism should be incorporated within the clutch to eliminate noise produced in the transmission.

5. Size -

The size of the clutch must be smallest possible so that it should occupy minimum amount of space.

6. Inertia -

The clutch rotating parts should have minimum inertia otherwise when the clutch is released for gear changing, the clutch plate will causing hard shifting and gear clashing of the system.

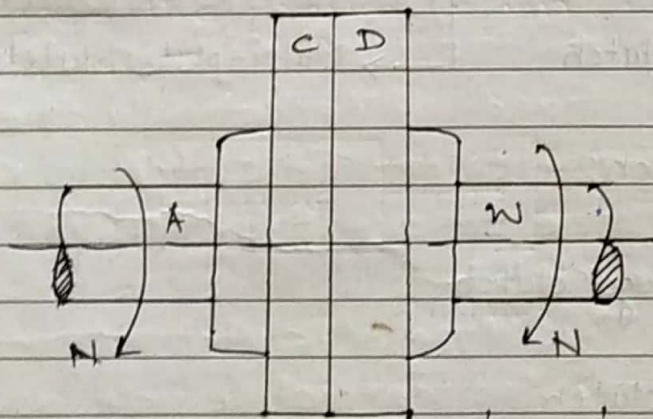
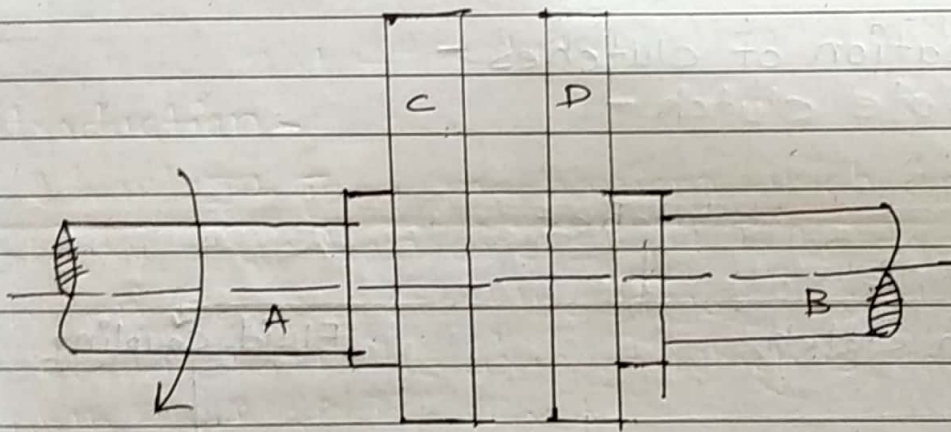
7. Ease of operation -

For higher torque transmission the operation of disengaging the clutch must not be tiresome to the driver.

* Function of clutch -

1. Engaging and Disengaging the gear box with the engine.
2. Transmitting the engine power to the road wheel smoothly.
3. Allowing the gear change without any damage.

* Principle of clutch -



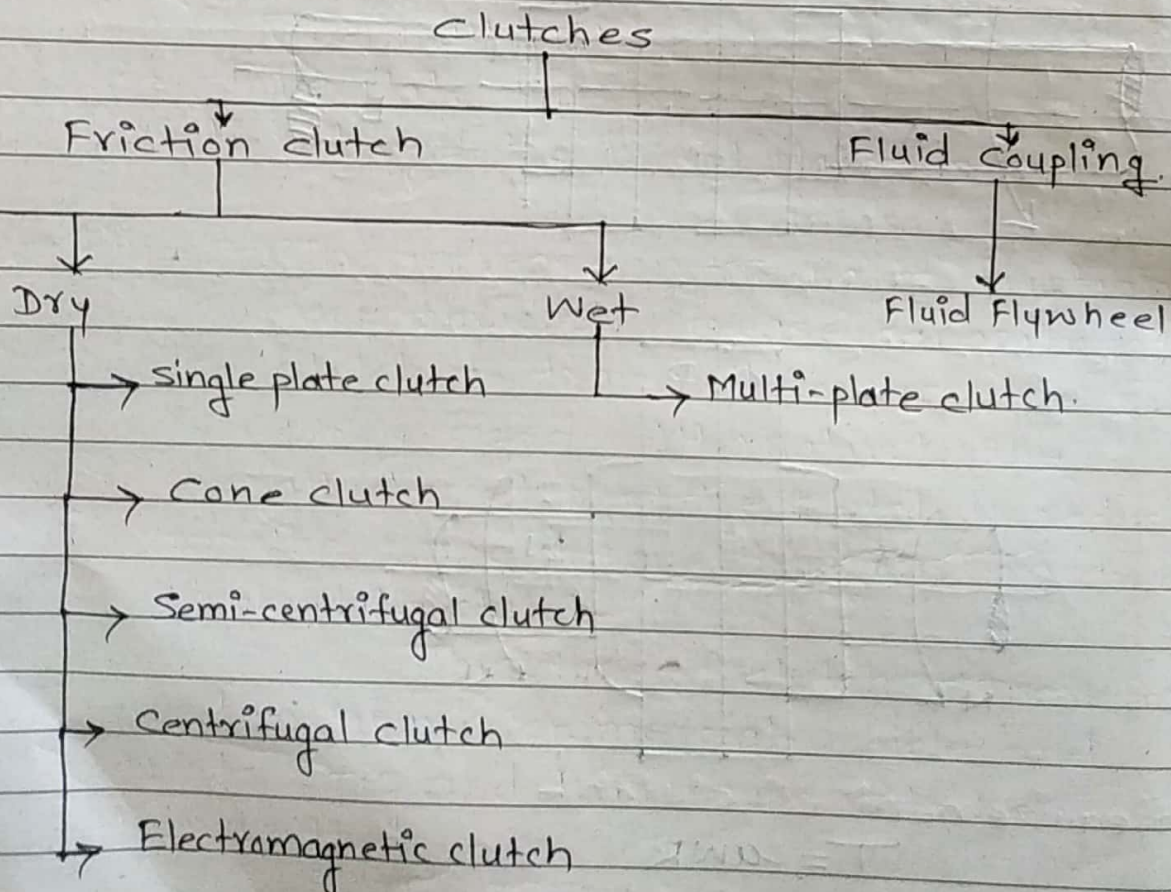
$$T = \mu W R$$

Where;

T - Torque transmitted

Let shaft A and disc 'c' be revolving at some speed (N rpm). shaft B and the disc 'D' fitted to keyed are stationary, initially when the clutch is not engaged (Fig. a). Now apply some axial force (w) to the disc D so that it comes in contact with disc 'c' as soon as the contact is made, the force of friction between 'c' and 'D' will come into play and frequently the disc 'D' will also start revolving, the speed of D depends upon friction force present, which in turn is proportional to the force (w) applied.

* classification of clutches -



* Cone clutch -

In this type the contact surface are in the form of cones as shown in the fig. In the engaged position the male cone is fully inside the female cone so that the friction surface are in complete contact. This is done by means of spring which keep the male cone pressed all the time.

When the clutch is engaged, the torque is transmitted from the engine via the flywheel and the male cone to the splined gear box shaft. For disengaging the clutch the male cone is pulled out by means of the lever system operated through the clutch pedal there by separating the contact surfaces.

* Single plate clutch -

Friction plate is held betⁿ the flywheel and the pressure plate there are springs (depending upon design) arranged circumferentially, which provide axial force to keep the clutch in engaged position.

The friction plate is mounted on a hub which is splined from inside and is free to slide over the gear box shaft. It is attached to the friction plate on both sides to provide to annular friction surfaces.

For the transmission of power, a pedal is provided to pull the pressure plate against the spring force whenever it is required to be disengaged.

Advantages -

1. With the single plate clutch, gear changing is easier than with the cone clutch, because the pedal movement is less in this case.

Disadvantages -

As compared to cone clutch the springs have to be more stiffness and this means that greater force required to be applied by the driver while disengaging.

* Multi-plate clutch -

The multiplate clutch is an extension of single plate type where the no. of frictional and the metal plates is increased.

Working - The increase in the no. of friction surface obviously increase capacity of the clutch to transmit torque, the size remaining fixed. Alternatively the overall diameter of the clutch is reduced for the same torque transmission as a single plate clutch.

This type of clutch is therefore used in heavy commercial vehicle and sports cars and high torque is to be transmitted.

The construction is to similar that of single plate except that all the friction plates in this case.

* Centrifugal clutch -

In this springs are eliminated together and only the centrifugal force is used to apply the required pressure. for keeping the clutch in engaged position.

As the speed increases, the centrifugal ball in flyes at here by operating the bell crank lever B which presses the plate 'c'. This force is transmitted to the plate D by means of spring E. The plate 'D' containing friction, thereby engaging clutch. Spring a serves to keep the clutch disengage at low speed, the stop 'H' limits the amount of centrifugal force.

Advantages -

1. Simple construction
2. Less maintenance
3. Affordable
4. Automatic.

Limitation -

1. Slipping
2. Power loss
3. low capacity
4. Overheating.

* Gear Box *

Introduction :-

Gear box is a speed and torque changing device between the engine & driving wheels. It serves the following purposes in the transmission system.

- ✓ It exchanges engine power from greater torque and thus provides a mechanical advantages to drive the vehicle under differential cond.
- ✗ It exchanges forward motion for reverse mot.
- ✗ It provides a neutral position to disallow power flow to the rest of the power train.

The transmission (gear box) acts in accordance with the running conditions. When driving power is required, it reduces the engine speed and transmits stronger torque to the driving wheels & when high running speed is desired, it transmits high speed low torque to the wheels. In addition the transmission serves to reverse the vehicle.

Types of gear boxes :-

Following types of gearboxes are used in automobiles,

① Selective type

- i) sliding mesh
- iii) Synchromesh
- ii) Constant mesh

② Progressive type

③ Epicyclic or planetary type.

The gear box provides a way of varying the engine's output torque & speed to match the vehicle's speed & load.

In order to achieve a high maxi. vehicle speed, combined with good accⁿ & economy over the whole speed range, a gearing system is required, which permits engine to best performance.

Maxi. engine power, torque & economy all occur at different engine speeds. As a result it becomes difficult to match the gear ratio for best performance, especially when variable operating conditions & driver demands are also to be considered.

Selective type gear boxes :-

It is that transmission in which any gear may be selected from the neutral position. In this type of transmission, neutral position has to be obtained before selecting any forward or reverse position.

Advantages:

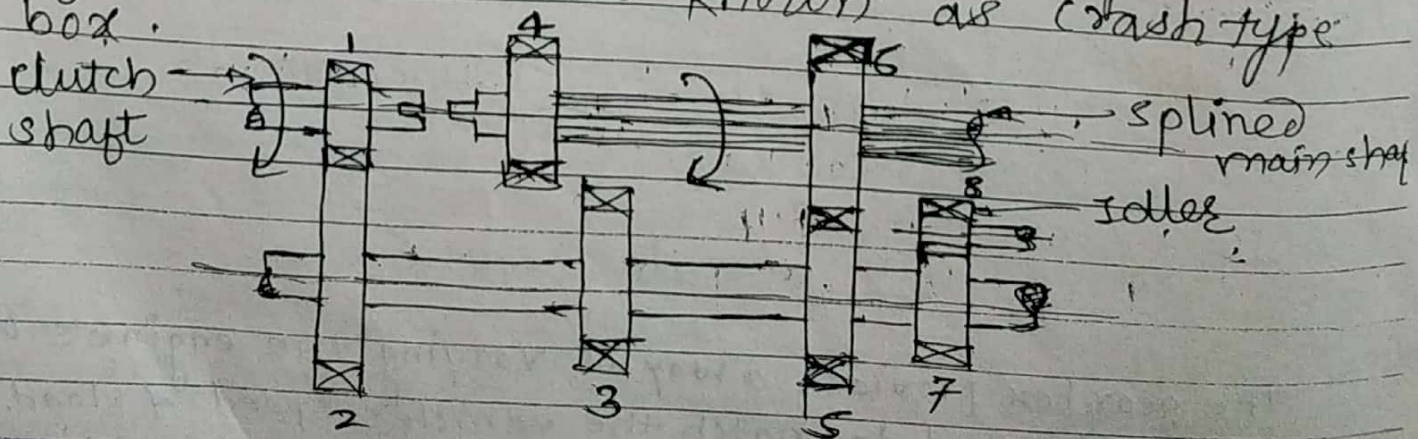
- Simple in construction
- Relatively free from troubles.
- Light & small
- Low production cost

Disadvantages:

- Gear ratios not being continuous but being in steps
- Noisy in operation.

@ Sliding mesh gear box :-

It is that gear box in which the gears on the splined main shaft are moved left or right for meshing them with appropriate gear on the layshaft for operating different speeds. These gear boxes are also known as crash type gear box.



Synchromesh - A system of gear changing, especially in motor vehicles, in which the driving & driven gearwheels are made to revolve at the same speed during engagement by means of a set of friction clutches, thereby easing the change.

This gear box consist of a clutch shaft, intermediate shaft and main shaft. The clutch shaft has one gear. The counter or lay shaft has four gears. All four gears form integral part of the counter shaft. The main shaft has two gears. These two gears can slide horizontally along the splines of the main shaft. At the same time the gears rotate the main shaft. There is no such splined arrangement in the countershaft, as such the gears cannot move along it.

First gear: (Fig 7.18)

When the gear shift lever is applied such that gear '6' meshes with gear '5', gear '4' doesn't mesh with gear '3'.

- When clutch shaft is rotating, drive is transmitted from gear '1' to gear '2'. Thus the countershaft is also rotated. The direction of rotating of countershaft is opposite to that of clutch shaft.
- When countershaft is rotating, gear '5' rotates gear '6', therefore gear '6' rotates the main shaft.

Thus the drive is transmitted along gears '1', '2', '5', and '6'. Gear '1' is smaller than gear '2', and gear '5' is smaller than gear '6'.

Second gear

When the clutch shaft is rotating, the transmission takes place between gears '1' & '2', '3' & '4'. Gear '4' rotates the main shaft. as shown in fig (4.18)

It having a system of clutches that synchronizes the speeds of the driving & driven members before engagement to avoid shock in gear changing & to reduce noise & wear.

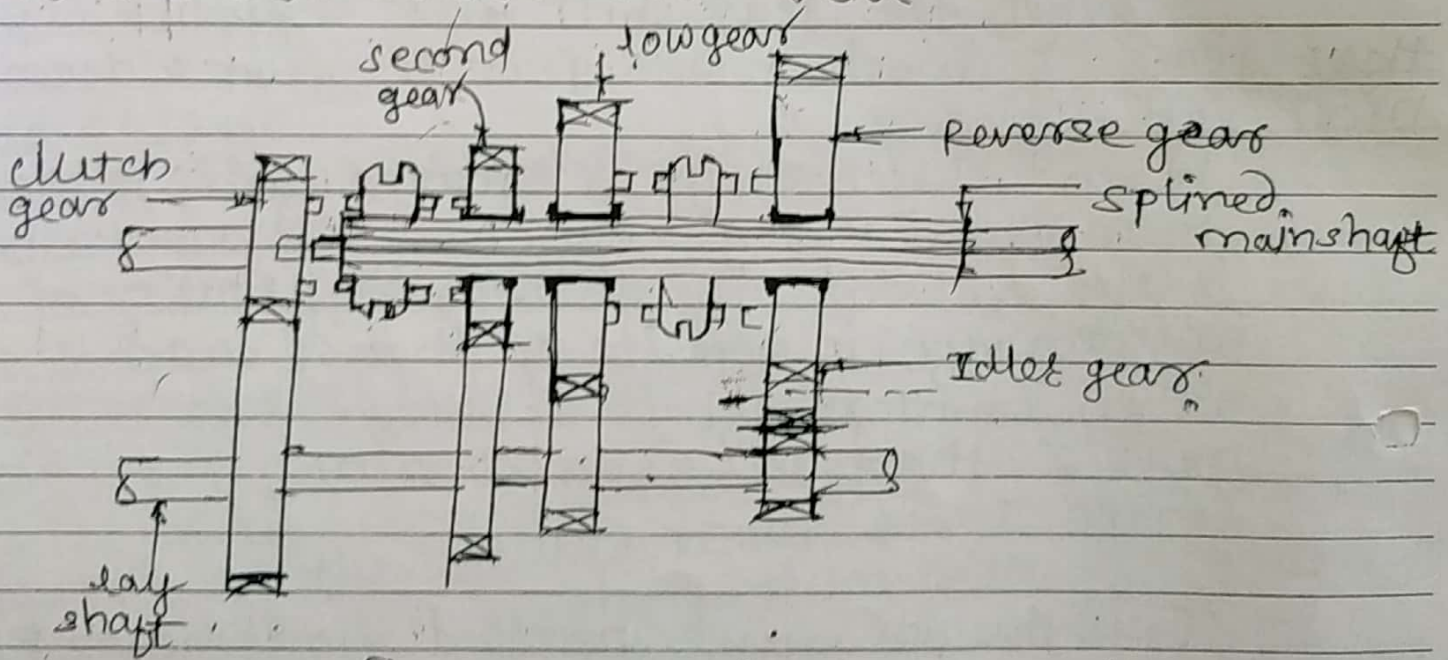
Third gear :-

As shown in fig, the dog of gear '4' is in direct mesh with gear '1'. Main shaft is in direct contact with the clutch shaft. Therefore, the main shaft acquires the same speed as clutch shaft.

Reverse gear :-

In the reverse gear transmission takes place from gear '1' to '2', '7' to '8' & then to '6'.

b) Constant mesh gear box :-



See fig 7.23. It is that gear box in which the gearbox gears are in 'constant mesh' with each other all the time & this gives a silent or quiet operation. Helical gears are used to make gear changing easier. The gears on main shaft which is splined, are free. The gears on countershaft are fixed. Two dog clutches provide in main shaft one between the clutch gear & second gear and other between low/first gear & reverse gear.

clutch can slide on the main shaft and rotate with it.

If the left hand dog clutch is made to slide to left by means of gearshift lever, it meshes with the 1st gear and the top speed gear is obtained. When the clutch meshes with the second gear, the second speed is obtained. Similarly by sliding the right-hand dog clutch to left & right, the first speed gear & reverse gear speed are obtained respectively. However, careful handling is necessary on the part of the driver that the speed of the locking dogs and respective rotation remain same to effect a clash free gear change.

Advantage :-

- 1) free from clashing
- 2) Safe from being damaged
- 3) Unpleasant grinding sound doesn't occur.

Disadvantage:

- 1) more defects than synchromesh type
- 2) Necessity of double declutching.

1) Synchromesh gear box :-

It is that gear box in which sliding synchromeshing unit is provided in place of sliding dog clutches as in case of constant mesh gear box. With the help of synchronising unit, speed of both the driving & driven shaft is synchronised before they are clutched. Power flows through train of gears. Power flow from various shafts remains same as in constant mesh gear box.

'Synchronesh gear box' work on the principle that two gears to be engaged are first brought into frictional contact which equalises their speed after which they are engaged readily and smoothly.

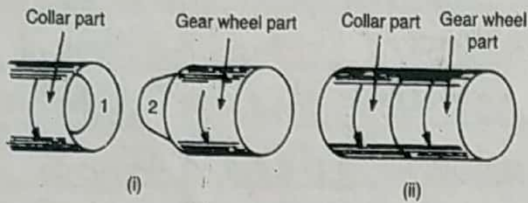


Fig. 7.24. Two cones as friction clutch.

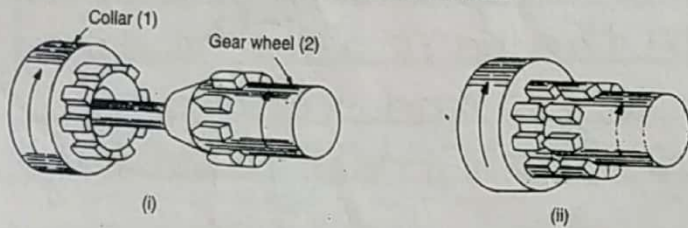


Fig. 7.25. Collar meshing with gear wheel.

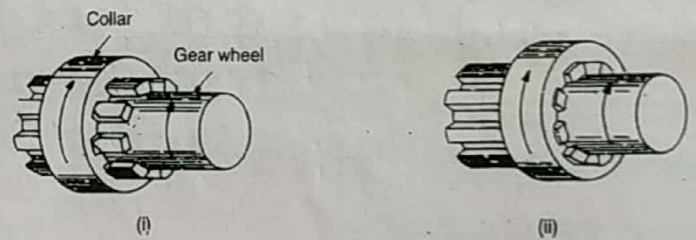


Fig. 7.26. Collar and gear wheel revolving together.

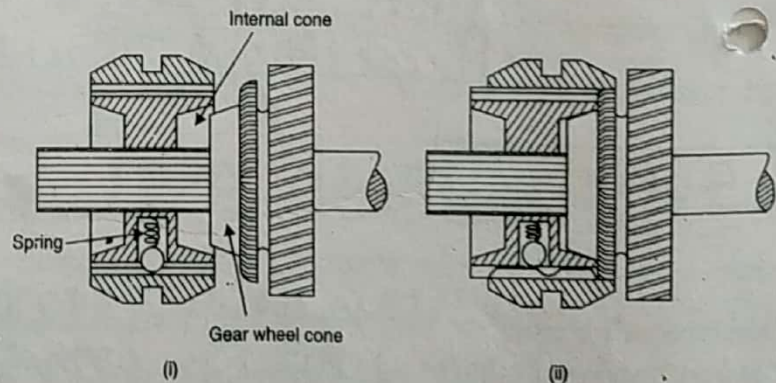


Fig. 7.27. Synchronesh unit.

Fig 7.24 (i) shows two conical surfaces, one-1 is the part of collar & cone-2 is part of gear wheel. cone 1 & 2 are revolving at different speeds. While cone-2 is revolving cone 1 gradually slides into it. Finally both cones revolve at same speed fig 7.24(ii)



Refer to fig 7.25(i), collar '1' & gear wheel '2' as separate and revolving at different speed. In fig 7.25(ii) the internal cone of collar comes in contact with the outer cone of gear wheel. In fig 7.27(i) the collar & gear wheel are shown close to each other. The cone of gear wheel and internal cone of the collar are in contact. The collar acquires the same speed as gear wheel. In fig 7.27(ii)

Advantages:

The synchromesh type of transmission has the big advantage of allowing smooth & quick shifting of gears without danger of damaging the gears & with necessity for double clutching.

① Epicyclic or Planetary gear box :-

This type of gear box uses no sliding dogs or gears to engage but different gear speeds are obtained by merely tightening brake-bands. Planetary gear set (fig 7.28) consists of a ring gear, annular wheel, sun gear & planet gears with carrier in order to obtain different speeds.

The ring gear contains teeth on its inner circumference & is surrounded by brake band.



Sun Gear → Planet Gear → Ring Gear

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A of 2 around

- The sun gear is rotated by the driving shaft from the engine & thus moves along with the movement of engine crankshaft.

- The planet gears are in constant mesh with both the sun gears and ring gear are free to rotate.

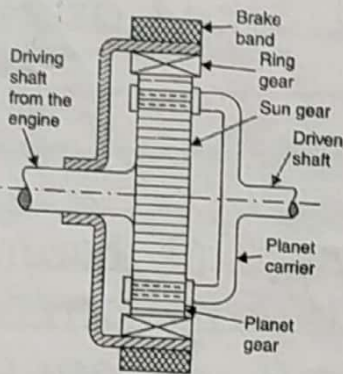


Fig. 7.28. Epicyclic gear set.

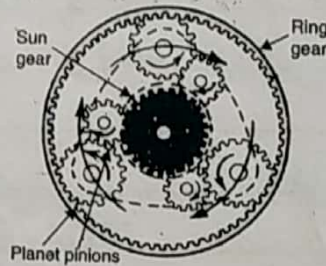


Fig. 7.29. Idler gear fitted in planetary system for reversing direction of rotation.

Epicyclic Gears are very widely used in automatic transmission because;

- They are always in constant Mesh
- engagement of these gears may be obtained smoothly

Working :-

- When the ring gear is locked by the brake band, rotating sun gear cause the planet gears to rotate. Since the ring gear cannot move, the planet gears are forced to climb over it. During this position ring gear acts as a track for the planet gears to move over.
- The driven shaft which is connected to planet gear carrier is thus rotated.

Three types of gears are used for final drive gearing

- a) straight bevel gears
- b) spiral bevel gears
- c) Hypoid gears.

Advantages:

- 1) High reduction ratio in single reduction, as high as 15:1
- 2) High ground clearance - when worn is over sprung.
- 3) Runs silently.

Disadvantages:

- 1) High cost of mfg
- 2) Low efficiency
- 3) Heavy demand on lubricants.

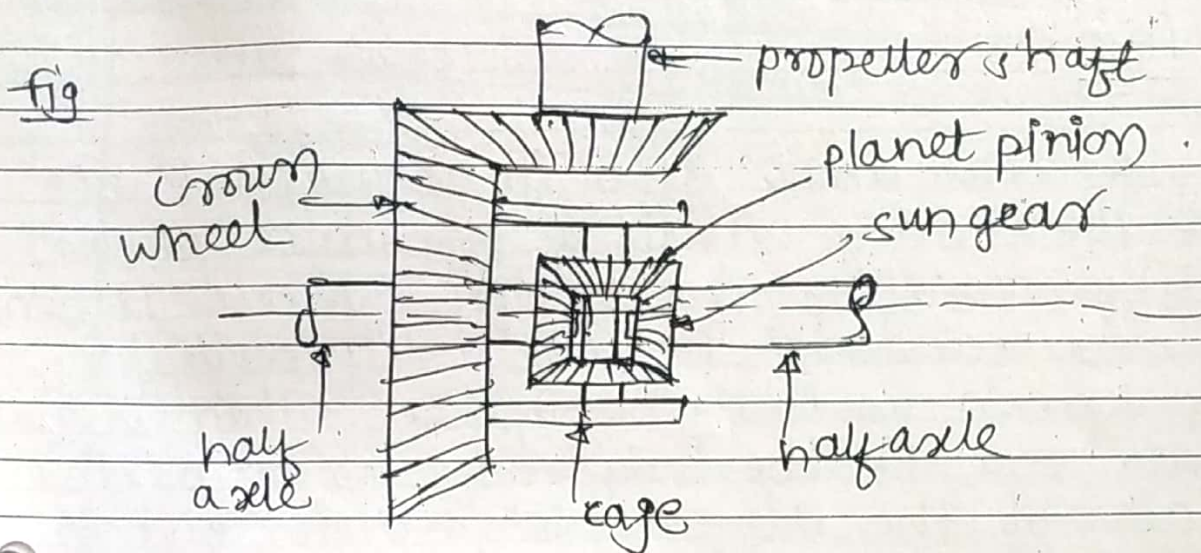
EMP

Differential :-

Differential is the mechanism by means of which outer wheels run faster than the inner wheels while taking a turn or moving over uneven road.

The purpose of the differential is to provide the relative motion to the two rear wheels when car is taking a turn.

Differential are used in rear drive as well as front engine, rear wheel drive vehicles. So four wheel drive vehicles having two differentials.



Construction:

fig shows various parts of differential unit. The bevel pinion is fixed to the propeller shaft which rotates the crown wheel. The crown wheel has another unit called the differential unit. It consists of two gears (sun gears) & two bevel gears (planet pinions).

Operation:

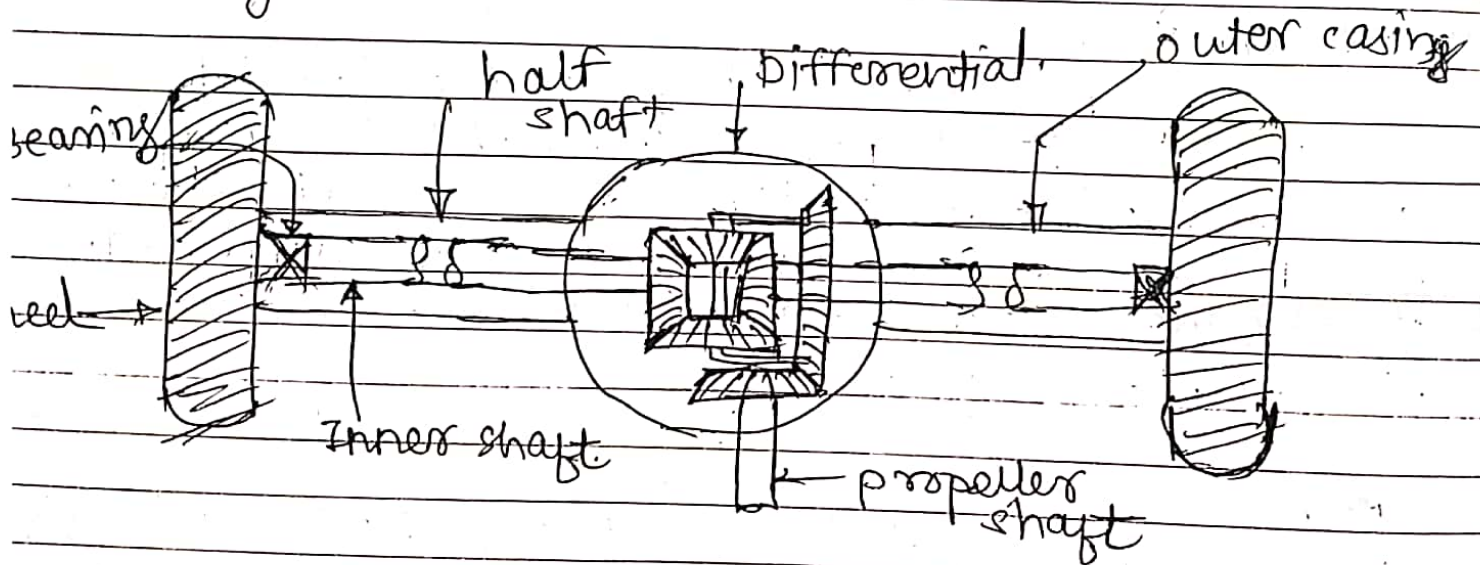
When the vehicle is moving on the straight path, ring gear, cage, crown wheel, sun gears rotate as a unit. Only two planet gears do not rotate on the pinion shaft.

When the car begins to round a curve, differential pinion gears i.e. planet gears rotate on the pinion shaft. This permits the outer wheel to turn faster than the inner wheel.

VIMP

Rear axles $\Rightarrow$

A rear axle lies in between the differential & the driving wheels & transmits power from the differential to the driving wheels. It consists of two halves connected by the differential, one part is known as half shaft & other is half shaft. Inner end of the half shaft is connected to the sun gear of the differential, & outer end to the driving wheel.



Rear axle performs following two functions:

- 1) Carry weight of vehicle
- 2) To rotate the wheel & transmit power from engine to wheels.

Almost all rear axle on modern passenger cars are live axle, that is they revolve with the wheels. Dead axle simply remains stationary. Housing or casing simply encloses the axle & differential, protects them from water, dust & injury.